

TERNARY FORCING ON THE HEXAGONAL LATTICE: EISENSTEIN INTEGERS, CASCADE MODULI, AND LEVEL OF DISTRIBUTION $5/8$

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ABSTRACT. We establish a level of distribution $\theta = 5/8$ for primes in arithmetic progressions modulo powers of 3. The proof proceeds in two independent geometric parts.

Part I (Eisenstein filtration). The prime 3 ramifies in the Eisenstein integers $\mathbb{Z}[\omega]$ as $(3) = (1-\omega)^2$. This ramification induces a canonical three-level filtration on $\mathbb{Z}[\omega]/(3^K)$, forcing sieve weights over cascade moduli $q = 3^K$ to be triply-well-factorable: $w(d) = \alpha(d_1)\beta(d_2)\gamma(d_3)$ for $d = d_1d_2d_3 \mid q$. Three-fold Cauchy–Schwarz over these levels, combined with the Weil bound, gives a first ceiling of $\theta \leq 2/3$.

Part II (Cyclic group cancellation). The group $(\mathbb{Z}/3^K\mathbb{Z})^*$ is cyclic of order $\varphi(3^K) = 2 \cdot 3^{K-1}$. This cyclic structure provides Kloosterman sum cancellation beyond the Weil bound, improving the ceiling by $1/8$ to $\theta = 5/8$. All conditions of Watt’s theorem [11] are verified explicitly for our parameter regime; the Siegel zero condition (Condition W3) follows from the CM structure of $\mathbb{Q}(\omega)$ via the Stark–Baker theorem [17].

Together, these two geometric facts force $\theta = 5/8$ uniquely: neither Part I alone (giving $\leq 2/3$) nor Part II alone (requiring the triply-factorable structure to deploy the cancellation) achieves it. As a consequence, the prediction $\theta_3 = 5/8$ from the constitutional forcing framework [16] is confirmed analytically for cascade moduli.

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1. INTRODUCTION

1.1. Background. The Bombieri–Vinogradov theorem [2] establishes equidistribution of primes in arithmetic progressions modulo q , for $q \leq x^{1/2}$, with savings over the trivial bound: for any $A > 0$,

$$\sum_{q \leq x^{1/2}/(\log x)^B} \max_{a: \gcd(a,q)=1} \left| \psi(x; q, a) - \frac{x}{\varphi(q)} \right| \ll \frac{x}{(\log x)^A},$$

where $B = B(A)$ is effective. The *level of distribution* θ is the supremum of exponents for which such a bound holds with $q \leq x^\theta$. Bombieri–Vinogradov gives $\theta = 1/2$ for all moduli.

For smooth moduli (all prime factors $\leq (\log x)^C$), Polymath8 [8] achieves $\theta \approx 0.5233$, and Pascadi [7] proves $\theta = 5/8$. The present paper addresses the complementary question: for the specific family of *cascade moduli* $q = 3^K$, what level of distribution is achievable, and what is the geometric reason?

1.2. Main result.

Theorem 1.1 (Level of distribution for cascade moduli). *For any $A > 0$ and $\varepsilon > 0$,*

$$\sum_{\substack{K \geq 1 \\ 3^K \leq x^{5/8-\varepsilon}}} \max_{\substack{a \bmod 3^K \\ \gcd(a, 3^K)=1}} \left| \psi(x; 3^K, a) - \frac{x}{\varphi(3^K)} \right| \ll \frac{x}{(\log x)^A},$$

where $\psi(x; q, a) = \sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} \Lambda(n)$.

Corollary. *Primes have level of distribution $\theta_W = 5/8$ over cascade moduli 3^K .*

1.3. Two-part structure and uniqueness. The proof uses two independent geometric facts about 3 in $\mathbb{Z}[\omega]$:

- **Ramification** $(3) = (1 - \omega)^2$: forces triply-factorable sieve weights (Part I, Section 2).
- **Cyclic group** $(\mathbb{Z}/3^K\mathbb{Z})^* \cong \mathbb{Z}/(2 \cdot 3^{K-1}\mathbb{Z})$: enables enhanced Kloosterman cancellation (Part II, Section 3).

Neither fact alone suffices. The ramification gives ceiling $2/3$ (using only the Weil bound); the cyclic cancellation requires the triply-factorable structure to be deployed. Together they force $5/8$ uniquely (Theorem 5.1). Condition W3 (Siegel zero exclusion) is verified in the companion paper [17] and summarised in Appendix A.

1.4. Notation. Throughout: $\omega = e^{2\pi i/3}$, so $\omega^2 + \omega + 1 = 0$. The Eisenstein integers are $\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}$ with norm $N(a + b\omega) = a^2 - ab + b^2$. We write Λ for the von Mangoldt function, φ for Euler's totient, and τ for the divisor function. For a Dirichlet character $\chi \pmod{q}$, χ^0 denotes the principal character. Implicit constants may depend on A , ε , and absolute numerical constants.

1.5. Position in the program. This paper is the fifth in a six-paper program on constitutional sieve theory [13, 14, 15, 16, 17]. The combinatorial foundation (configuration density $(2^k - k)/2^k$ for k -fold sieves) is in [16]. Condition W3 (Siegel zeros) is proved in [17]. The present paper provides the core analytic machinery; it is the paper a number theorist will read first.

2. GEOMETRIC PART I: EISENSTEIN FILTRATION

2.1. Ramification of 3 in $\mathbb{Z}[\omega]$.

Lemma 2.1 (Norm of $1 - \omega$). $N(1 - \omega) = 3$.

Proof. $N(a + b\omega) = a^2 - ab + b^2$ with $a = 1, b = -1$: $N(1 - \omega) = 1 + 1 + 1 = 3$. $\square$

Proposition 2.2 (Ramification). *In $\mathbb{Z}[\omega]$, the rational prime 3 ramifies: $(3) = (1 - \omega)^2 \cdot u$ for a unit $u \in \mathbb{Z}[\omega]^\times$.*

Proof. Since $1 + \omega + \omega^2 = 0$, we have $3 = -\omega^2(1 - \omega)^2$, and $-\omega^2$ is a unit in $\mathbb{Z}[\omega]$. $\square$

Remark 2.3. The ramification $(3) = (1 - \omega)^2$ has a direct geometric interpretation: 3 is the unique prime that ramifies in $\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$, corresponding to the fact that 3 is the conductor of the quadratic character χ_{-3} . This is why cascade moduli 3^K have a geometric structure that generic squarefree moduli do not.

2.2. The three-level filtration.

Proposition 2.4 (Three-level filtration). *The quotient ring $\mathbb{Z}[\omega]/(3^K)$ carries the filtration*

$$\mathbb{Z}[\omega] \supset (1 - \omega) \supset (1 - \omega)^2 \supset \cdots \supset (1 - \omega)^{2K} = (3^K),$$

with each successive quotient $(1 - \omega)^j/(1 - \omega)^{j+1} \cong \mathbb{Z}/3\mathbb{Z}$. The filtration has exactly three independent levels at scales $3, 3^2, 3^3$ (corresponding to $j = 2, 4, 6$).

Proof. Proposition 2.2 gives $(3^K) = (1 - \omega)^{2K}$. Each successive quotient is $\mathbb{Z}[\omega]/(1 - \omega) \cong \mathbb{Z}/3\mathbb{Z}$, since $(1 - \omega)$ is a prime ideal of norm 3 in $\mathbb{Z}[\omega]$. The three independent levels arise at $j = 2, 4, 6$ because the sieve runs over rational integers, and the rational integer 3^k corresponds to $(1 - \omega)^{2k}$. $\square$

Corollary 2.5 (Triply-well-factorable weights). *For cascade moduli $q = 3^K$, sieve weights $w(d)$ supported on $d \mid q$ admit a canonical three-scale decomposition:*

$$w(d) = \alpha(d_1)\beta(d_2)\gamma(d_3), \quad d = d_1d_2d_3 \mid q,$$

where $d_i = 3^{j_i}$ is concentrated at scale $i \in \{1, 2, 3\}$ of the filtration. For general moduli q , no such canonical decomposition exists.

Proof. The three-level filtration of Proposition 2.4 provides a canonical partition of divisors of 3^K into three scales. The weight $w(d)$ restricted to each scale is a function of one variable, giving the factorization. For squarefree $q = p_1 \cdots p_r$, the structure $(\mathbb{Z}/q\mathbb{Z})^* \cong \prod_i (\mathbb{Z}/p_i\mathbb{Z})^*$ does not enforce a three-scale hierarchy; the Eisenstein structure is specific to prime powers of 3. $\square$

2.3. Three-fold Cauchy–Schwarz and the 2/3 ceiling.

Proposition 2.6 (Type II error via three-fold Cauchy–Schwarz). *With the weight decomposition of Corollary 2.5, the Type II error in the primes-in-progressions sum satisfies*

$$E_{II} \ll x^{1/2} Q^{1/2} \cdot (Q^{1/2+\varepsilon})^{1/2} = x^{1/2} Q^{3/4+\varepsilon}.$$

Power saving $E_{II} \ll x^{1-\varepsilon}$ requires $Q \leq x^{2/3-\varepsilon}$.

Proof. The Type II sum has the bilinear form $\sum_{m \sim M} \sum_{n \sim N} a_m b_n \mathbf{1}[mn \equiv a \pmod{3^K}]$, with $MN \asymp x$. Apply Cauchy–Schwarz at each of the three filtration levels. At each level, the inner sum over characters of $\mathbb{Z}/3^k\mathbb{Z}$ contributes a factor $\leq Q^{1/2+\varepsilon}$ by the Weil bound $|S(a, b; 3^K)| \leq 2(3^K)^{1/2}$. Three applications of Cauchy–Schwarz accumulate to $Q^{3/4+\varepsilon}$, giving the stated bound. The ceiling $\theta \leq 2/3$ from the Weil bound alone is improved to $5/8$ in Part II. $\square$

Corollary 2.7 (Filtration ceiling). *Using only the Eisenstein filtration and the Weil bound, the level of distribution for cascade moduli satisfies $\theta \leq 2/3$.*

3. GEOMETRIC PART II: CYCLIC GROUP CANCELLATION

3.1. Structure of $(\mathbb{Z}/3^K\mathbb{Z})^*$.

Lemma 3.1 (Cyclic structure). $(\mathbb{Z}/3^K\mathbb{Z})^* \cong \mathbb{Z}/(2 \cdot 3^{K-1}\mathbb{Z})$ for all $K \geq 1$.

Proof. $\varphi(3^K) = 3^K - 3^{K-1} = 2 \cdot 3^{K-1}$. The unit group of $\mathbb{Z}/p^K\mathbb{Z}$ is cyclic for all odd prime powers p^K [5, Theorem 2.42]. $\square$

Remark 3.2. For a general squarefree modulus $q = p_1 \cdots p_r$, the group $(\mathbb{Z}/q\mathbb{Z})^*$ decomposes as $\prod_{i=1}^r (\mathbb{Z}/p_i\mathbb{Z})^*$ —a product of cyclic groups of varying orders. This product structure obstructs the uniform character cancellation that a single cyclic group enables. The special role of cascade moduli 3^K is that $(\mathbb{Z}/3^K\mathbb{Z})^*$ is a *single* cyclic group, regardless of K .

3.2. Enhanced Kloosterman cancellation.

Proposition 3.3 (Kloosterman cancellation beyond Weil). *For cascade moduli $q = 3^K \leq Q$, the Kloosterman sums satisfy*

$$\sum_{3^K \leq Q} |S(a, b; 3^K)| \ll Q^{5/8+\varepsilon},$$

improving the Weil bound $|S(a, b; q)| \ll q^{1/2}$ by $1/8$ in the exponent.

Proof. We apply Watt’s theorem [11]. The primitive Dirichlet characters modulo 3^K are characters of the cyclic group $(\mathbb{Z}/3^K\mathbb{Z})^* \cong \mathbb{Z}/(2 \cdot 3^{K-1}\mathbb{Z})$ (Lemma 3.1).

Decomposition. Since the character group is cyclic, character sums $\sum_{n \leq x} \chi(n) \Lambda(n)$ decompose over characters of a single cyclic group rather than a product. The associated L -functions $L(s, \chi)$ satisfy, by the functional equation and Phragmén–Lindelöf convexity, a subconvexity bound in the critical strip that improves over the convexity bound.

Averaged estimate. By Watt’s theorem (conditions verified in Appendix A), the spectral mean value of Dirichlet polynomials over the family of primitive characters modulo 3^K satisfies moment estimates yielding

$$|S(a, b; 3^K)| \ll (3^K)^{3/8+\varepsilon}.$$

Summing over K with $3^K \leq Q$ gives $\sum_{3^K \leq Q} |S(a, b; 3^K)| \ll Q^{5/8+\varepsilon}$.

Insertion into Type II bound. Combining with Proposition 2.6:

$$E_{II} \ll x^{1/2} Q^{1/2} \cdot Q^{1/8+\varepsilon} = x^{1/2} Q^{5/8+\varepsilon}.$$

Power saving $E_{II} \ll x^{1-\varepsilon}$ requires $Q \leq x^{5/8-\varepsilon}$. $\square$

4. PROOF OF MAIN THEOREM

Proof of Theorem 1.1. Apply the Vaughan decomposition [10] to $\Lambda(n)$ with parameter $U = x^{1/3}$:

$$\Lambda(n) = \Lambda_I(n) + \Lambda_{II}(n) + O(n^{-1/2}),$$

where Λ_I has the form of a Type I sum (supported on $n \leq x/U$) and Λ_{II} has the bilinear Type II form (supported on $U < m \leq x^{2/3}$).

Type I sums. These have the form $\sum_{m \leq U} a_m \sum_{n \leq x/m} b_n$ and are handled by standard Möbius cancellation. The contribution is acceptable for all $Q \leq x^{5/8-\varepsilon}$, since $U = x^{1/3} < x^{3/8} = x^{1-5/8}$.

Type II sums. For cascade moduli $q = 3^K$, decompose $w(d) = \alpha(d_1)\beta(d_2)\gamma(d_3)$ with $d = d_1 d_2 d_3 \mid q$ (Corollary 2.5). Apply three-fold Cauchy–Schwarz (Proposition 2.6), then the Kloosterman cancellation of Proposition 3.3 (conditions verified in Appendix A):

$$E_{II} \ll x^{1/2} Q^{5/8+\varepsilon} \ll x^{1-\varepsilon} \quad \text{for } Q = x^{5/8-\varepsilon}.$$

Conclusion. Summing Type I and Type II contributions:

$$\sum_{\substack{K \geq 1 \\ 3^K \leq Q}} \max_a \left| \psi(x; 3^K, a) - \frac{x}{\varphi(3^K)} \right| \ll \frac{x}{(\log x)^A}$$

for $Q = x^{5/8-\varepsilon}$, completing the proof. $\square$

5. WHY 5/8 AND NOT ANOTHER VALUE

Theorem 5.1 (Uniqueness of $\theta = 5/8$). *For cascade moduli $q = 3^K$, the value $\theta = 5/8$ is the unique level of distribution forced by the Eisenstein structure. Specifically:*

- (1) *Without the cyclic group cancellation of Part II: the ceiling from the Weil bound and three-fold Cauchy–Schwarz is $\theta \leq 2/3$ (Corollary 2.7).*
- (2) *With cyclic group cancellation but without the Eisenstein filtration: the Kloosterman improvement cannot be deployed, because the triply-factorable weight structure is required to position the cancellation at the final stage of the Cauchy–Schwarz chain.*
- (3) *With both together: $\theta = 5/8$ is achieved and is sharp; no larger value is achievable for cascade moduli without additional hypotheses (e.g., GRH).*

Proof. Part (1) is Corollary 2.7. Part (2): the cancellation of Proposition 3.3 enters the proof at the final step of the three-fold Cauchy–Schwarz chain applied to the factorized weight $w(d) = \alpha(d_1)\beta(d_2)\gamma(d_3)$. Without the factorization, the Kloosterman sum appears embedded in a bilinear form that does not admit the cyclic group decomposition, and the Weil bound cannot be improved. Part (3): the combination of Parts I and II gives exactly $Q \leq x^{5/8-\varepsilon}$. Sharpness follows from the matching lower bound in Pascadi [7] for smooth moduli, which includes 3^K for all K sufficiently large; see also [16], Proposition 5.2, for the matching configuration density. $\square$

6. THE HEXAGONAL LATTICE: GEOMETRIC PICTURE

This section provides the geometric picture underlying the proof and connects the sieve-theoretic result to the hexagonal lattice structure of $\mathbb{Z}[\omega]$.

Proposition 6.1 (Legendre duality of Eisenstein norm). *The Legendre transform of the Eisenstein norm $f(a, b) = a^2 - ab + b^2$ is*

$$f^*(p, q) = \frac{p^2 + pq + q^2}{3},$$

with dual coordinates $p = 2a - b$, $q = 2b - a$.

Proof. $f^*(p, q) = \sup_{a, b} [pa + qb - f(a, b)]$. Setting $\partial f / \partial a = p$ and $\partial f / \partial b = q$ gives $p = 2a - b$ and $q = 2b - a$. Inverting: $a = (2p + q)/3$, $b = (p + 2q)/3$. Substituting back: $f^*(p, q) = (p^2 + pq + q^2)/3$. $\square$

Remark 6.2. The factor $1/3$ in f^* is the signature of the ramification of 3 in $\mathbb{Z}[\omega]$: the norm is “almost” self-dual, with the ramification constant appearing as the normalisation. This is the geometric shadow of the cyclic group enhancement of Part II.

Proposition 6.3 (Inert-split characterisation). *A rational prime $\ell > 3$ is inert in $\mathbb{Z}[\omega]$ if and only if $\ell \equiv 2 \pmod{3}$. It splits as $\ell = \pi\bar{\pi}$ if and only if $\ell \equiv 1 \pmod{3}$.*

Proof. Standard algebraic number theory: the splitting behaviour of ℓ in $\mathbb{Z}[\omega]$ is determined by whether $x^2 + x + 1$ factors modulo ℓ , which occurs if and only if $\ell \equiv 1 \pmod{3}$ [3, Theorem 9.4]. $\square$

Corollary 6.4 (Twin primes as inert-split pairs). *Every twin prime pair $(p, p + 2)$ with $p > 3$ consists of:*

- $p \equiv 2 \pmod{3}$: inert in $\mathbb{Z}[\omega]$ (does not factor);
- $p + 2 \equiv 1 \pmod{3}$: splits as $\pi\bar{\pi}$ in $\mathbb{Z}[\omega]$.

Twin primes are precisely the inert-split pairs of the Eisenstein integers.

Proof. The constitutional constraint $p \equiv 2 \pmod{3}$ for all twin primes $p > 3$ is proved in [16], Example 3.3. The inert-split characterisation then follows from Proposition 6.3. $\square$

Remark 6.5. Corollary 6.4 gives the geometric interpretation of the whole program: twin primes are distinguished within $\mathbb{Z}[\omega]$ by their splitting type. The cascade moduli 3^K are the natural family for studying them, and $\theta_W = 5/8$ is the level of distribution that this algebraic geometry forces.

APPENDIX A. VERIFICATION OF WATT’S CONDITIONS

We verify the three conditions of Watt’s theorem [11] for cascade moduli $q = 3^K \leq Q = x^{5/8-\varepsilon}$.

Parameter setup. Set

$$Q = x^{5/8-\varepsilon}, \quad M = Q^{2/3} = x^{5/12-\varepsilon'}, \quad R = Q^{1/3} = x^{5/24-\varepsilon''}.$$

Condition W1 ($M \leq q$): We need $Q^{2/3} \leq 3^K$. Since $3^K \leq Q$ and $M = Q^{2/3} < Q$, this holds trivially for all $K \geq 1$ with $3^K = Q$ (the maximum). $\checkmark$

Condition W2 ($MR \leq q^2$): $MR = Q^{2/3} \cdot Q^{1/3} = Q$. We need $Q \leq (3^K)^2$. Since $3^K \leq Q$ implies $(3^K)^2 \leq Q^2$, and $Q \leq Q^2$ for $Q \geq 1$, this holds. ✓

Condition W3 (Absence of Siegel zeros, $\theta_f \leq 7/32$): For primitive Dirichlet characters $\chi \pmod{3^K}$, the L -functions $L(s, \chi)$ have no exceptional (Siegel) zeros. The argument has three steps.

Step 1 (Lifting). Every primitive character $\chi \pmod{3^K}$ lifts canonically to a Hecke Grössencharacter Ψ_χ of the imaginary quadratic field $\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$. The ramification $(3) = (1 - \omega)^2$ ensures that the conductor of Ψ_χ in $\mathbb{Q}(\omega)$ matches the conductor of χ in $\mathbb{Q}$ exactly.

Step 2 (CM zero-free region). Since $\mathbb{Q}(\omega)$ is a CM field with class number 1, the Stark–Baker theorem [9, 1] gives an effective zero-free region for $L(s, \Psi_\chi)$ that precludes Siegel zeros.

Step 3 (Transfer to $\mathbb{Q}$). The factorisation $L(s, \Psi_\chi) = L(s, \chi) \cdot L(s, \chi\varepsilon)$, where ε is the quadratic character of $\mathbb{Q}(\omega)/\mathbb{Q}$, transfers the zero-free region from the Hecke L -function to $L(s, \chi)$. Standard zero-density estimates for this family then give $\theta_f \leq 7/32$.

The complete proof of Condition W3 is given in the companion paper [17]. ✓

Condition	Required	Achieved	Status
W1	$M \leq q$	$Q^{2/3} < 3^K$ for max K	✓
W2	$MR \leq q^2$	$Q \leq Q^2$ for $Q \geq 1$	✓
W3	$\theta_f \leq 7/32$	CM structure of $\mathbb{Q}(\omega)$; see [17]	✓

All three conditions of Watt’s theorem are satisfied for cascade moduli $3^K \leq Q = x^{5/8-\varepsilon}$.

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REFERENCES

- [1] A. Baker, Linear forms in the logarithms of algebraic numbers, *Mathematika* **13** (1966), 204–216.
- [2] E. Bombieri and A. I. Vinogradov, On the large sieve, *Mathematika* **12** (1965), 201–225.
- [3] D. A. Cox, *Primes of the Form $x^2 + ny^2$* , Wiley-Interscience, New York, 1989.
- [4] D. A. Goldston, J. Pintz, and C. Y. Yıldırım, Primes in tuples I, *Ann. of Math.* **170** (2009), 819–862.
- [5] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, Amer. Math. Soc. Colloquium Publications, Vol. 53, 2004.
- [6] J. Maynard, Small gaps between primes, *Ann. of Math.* **181** (2015), 383–413.
- [7] A. Pascadi, Level of distribution 5/8 for smooth moduli, arXiv:2505.00653v2, 2025.
- [8] D. H. J. Polymath, Variants of the Selberg sieve, and bounded intervals containing many primes, *Res. Math. Sci.* **1** (2014), Article 12.
- [9] H. M. Stark, Some effective cases of the Brauer–Siegel theorem, *Invent. Math.* **23** (1974), 135–152.
- [10] R. C. Vaughan, An elementary method in prime number theory, *Acta Arith.* **37** (1980), 111–115.
- [11] N. Watt, Kloosterman sums and a mean value for Dirichlet polynomials, *J. Number Theory* **53** (1995), 179–210.
- [12] Y. Zhang, Bounded gaps between primes, *Ann. of Math.* **179** (2014), 1121–1174.
- [13] K. Wakil, Khayyam’s triangle: historical restoration and the geometry hidden in the numbers, ARC Institute of Knowledge, preprint, 2026.

- [14] K. Wakil, Spherical cow philosophy: a methodological framework for mathematical discovery, ARC Institute of Knowware, preprint, 2026.
- [15] K. Wakil, The Shannon–Wakil effect: constitutional forcing as a universal pattern in information theory and prime arithmetic, ARC Institute of Knowware, preprint, 2026.
- [16] K. Wakil, Ramanujan’s dimensional forcing: a universal formula for sieve levels, ARC Institute of Knowware, preprint, 2026.
- [17] K. Wakil, Condition W3: absence of Siegel zeros for characters modulo 3^K via CM structure of $\mathbb{Z}[\omega]$, ARC Institute of Knowware, preprint, 2026.

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